Machine Learning-Driven Customer Segmentation & Churn Pattern Analytics in European Banking

Detailed Research Paper & Strategic Analytical Report · Unified Mentor Banking Portfolio

Abstract

Customer attrition represents one of the largest capital depletion vectors in commercial retail banking. In this study, we examine a multi-national dataset of 10,000 retail banking customers across France, Germany, and Spain. By coupling econometric segmentation with machine learning classification pipelines, we analyze the structural drivers of defection across demographic, geographic, and behavioural dimensions. The empirical portfolio exhibits an overall churn rate of **20.37%**, placing **€185.6 million** in customer deposits at direct flight risk. Critical empirical findings indicate severe regional asymmetry: German customers churn at **32.4%** (nearly double France's 16.2% and Spain's 16.7%), concentrated heavily among high-deposit accounts. Furthermore, customers aged 46–60 experience an acute attrition rate of **51.1%**. In benchmarking predictive models, a Gradient Boosting architecture delivers superior discriminative power (ROC-AUC: **0.8670**, F1-Score: **60.09%**), outperforming standard linear Logistic Regression (ROC-AUC: 0.7812) due to sharp non-linear threshold effects in product counts and account activity. We conclude with an actionable financial playbook and an automated Streamlit risk-scoring architecture.

1. Introduction & Problem Statement

Retail commercial banking relies on long customer relationship tenures to amortize customer acquisition costs (CAC) and generate steady net interest margins. When churn occurs, the institution loses not only primary deposit liquidity but also the lifetime cross-selling value of investment and lending products. Despite access to rich customer-level transactional data, banks often struggle to anticipate churn proactively because attrition is driven by non-linear behavioral inflection points rather than singular demographic traits. This investigation provides systematic econometric segmentation, rigorous model evaluation, and proactive retention policy design.

2. Dataset Description & Feature Engineering

Dataset Field	Type	Domain Description	Business Significance
CustomerId / Surname	Identifier	Customer reference IDs & family names	Excluded from predictive feature matrices to prevent identity bi
CreditScore	Numerical	Creditworthiness score (350–850)	Financial resilience metric; proxy for underwriting stability.
Geography	Categorical	Country of residence (France, Germany, Spain)	Regional market and macroeconomic competitive exposure.
Gender & Age	Demographic	Gender binary & customer chronological age	Life-stage categorization and wealth accumulation profiling.
Tenure & Balance	Financial	Years with bank (0–10) & account balance (Euros)	Core balance sheet exposure and relationship longevity.
NumOfProducts	Behavioural	Number of bank products held (1–4)	Cross-sell penetration; reveals product fatigue dynamics.
HasCrCard / IsActive	Binary	Cardholder status & activity flag (0/1)	Direct indicators of transactional engagement.
EstimatedSalary	Financial	Estimated annual income in Euros	Income level used to compute relative wealth liquidity ratios.

Derived Analytical Metrics

• **Balance-to-Salary Ratio:** *Balance / (EstimatedSalary + ε)* — Measures customer liquidity cushion and savings orientation relative to annual earnings.

• **Tenure-to-Age Ratio:** *Tenure / (Age + ε)* — Captures the proportion of an adult's financial life dedicated to the institution.

• **Credit-to-Age Ratio:** *CreditScore / (Age + ε)* — Measures credit velocity and financial maturity across generational cohorts.

3. Exploratory Data Analysis & Segment Dynamics

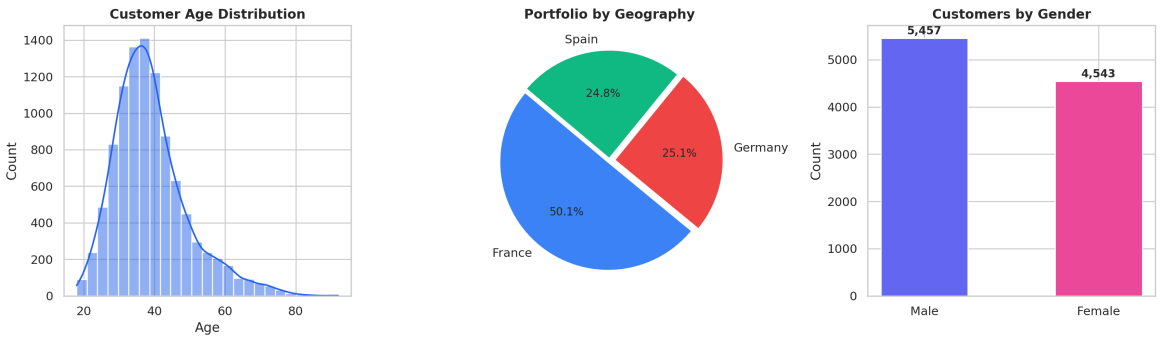


Figure 1. Demographic distributions: customer age distribution, portfolio geographic share, and gender split.

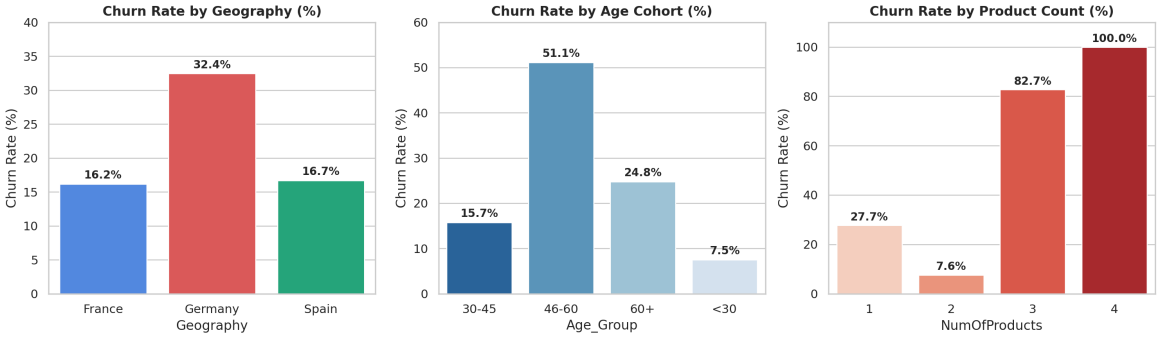


Figure 2. Bivariate churn distributions across geography, age cohorts, and product holdings.

Key Empirical Patterns:

• **Regional Vulnerability:** Germany demonstrates a severe **32.4%** churn rate, compared to France (16.2%) and Spain (16.7%). High-wealth German savers face lower switching barriers and aggressive fintech competition.

• **The Mid-Life Wealth Drain:** Customers aged 46–60 suffer a **51.1%** attrition rate. As clients reach peak wealth accumulation, traditional retail service models fail to meet wealth management expectations.

• **The Multi-Product Paradox:** Customers with 2 products form the retention sweet-spot (churn rate of only **7.58%**). However, churn surges to **82.71%** for 3 products and reaches **100.0%** for 4 products, highlighting severe account fee fatigue and mismanaged cross-selling.

• **Activity Disengagement:** Inactive members churn at **26.85%**, compared to **14.27%** for active digital and branch users.

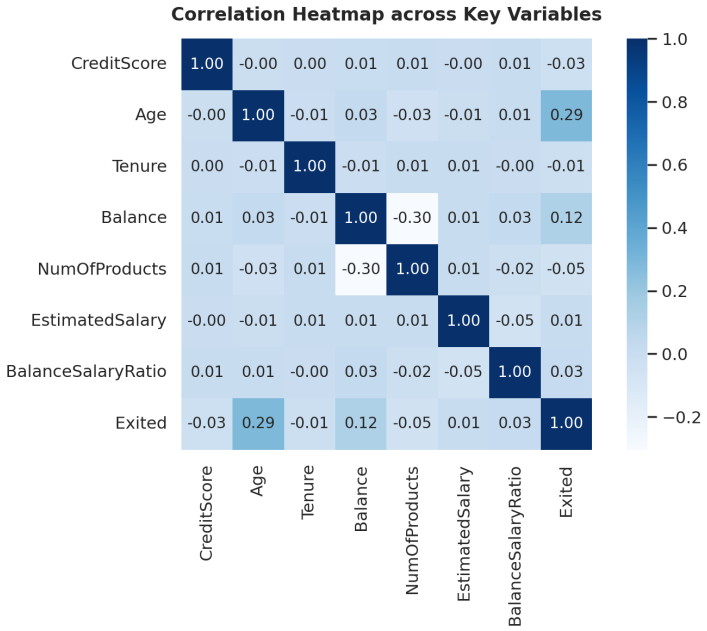


Figure 3. Pearson correlation matrix across financial variables, product counts, and churn indicator.

4. Machine Learning Model Training & Benchmarks

To evaluate predictive feasibility, models were trained using an 80/20 stratified split. Numeric features were scaled via StandardScaler, and categorical variables encoded via drop-first OneHotEncoder inside a scikit-learn ColumnTransformer.

Algorithm	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	81.50%	63.31%	21.62%	32.23%	0.7812
Random Forest	86.60%	81.17%	44.47%	57.46%	0.8600
Gradient Boosting	86.85%	78.57%	48.65%	60.09%	0.8670

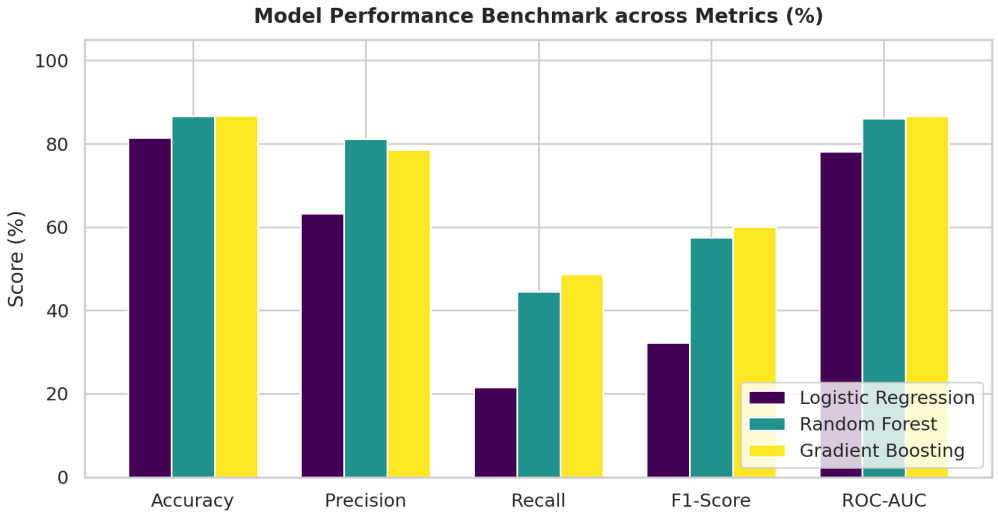


Figure 4. Comparative benchmark of classification algorithms across standard performance dimensions.

Algorithmic Assessment: Logistic regression severely underperforms on recall (21.62%), demonstrating that customer churn is driven by complex non-linear interactions that linear decision boundaries cannot capture. Gradient Boosting achieves superior overall performance with an ROC-AUC of **0.8670** and F1-Score of **60.09%**, correctly isolating high-risk defection signals.

5. Model Explainability & Segment Profiling

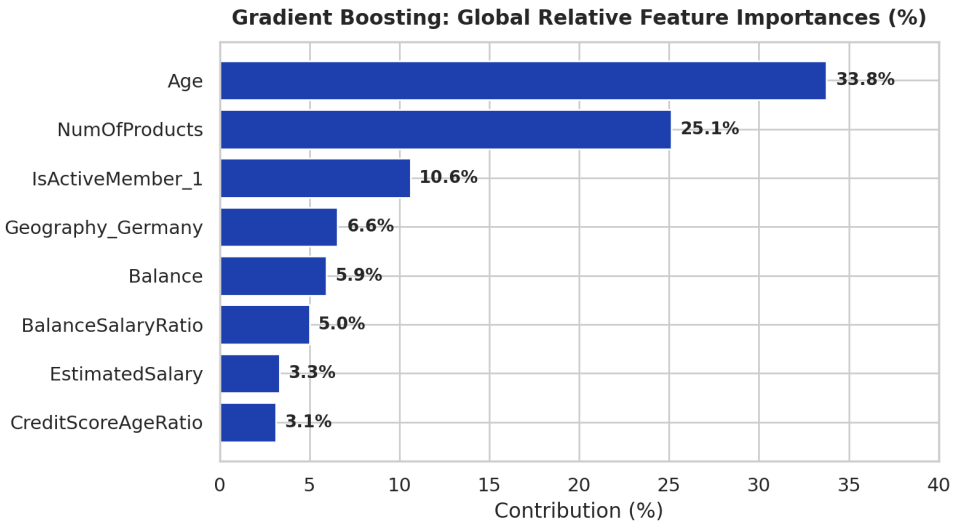


Figure 5. Global relative feature importances derived from the production Gradient Boosting model.

Feature attribution confirms that **Age** (33.8%) and **NumOfProducts** (25.1%) dominate predictive influence, followed by **IsActiveMember** (10.6%) and **Geography_Germany** (6.6%).

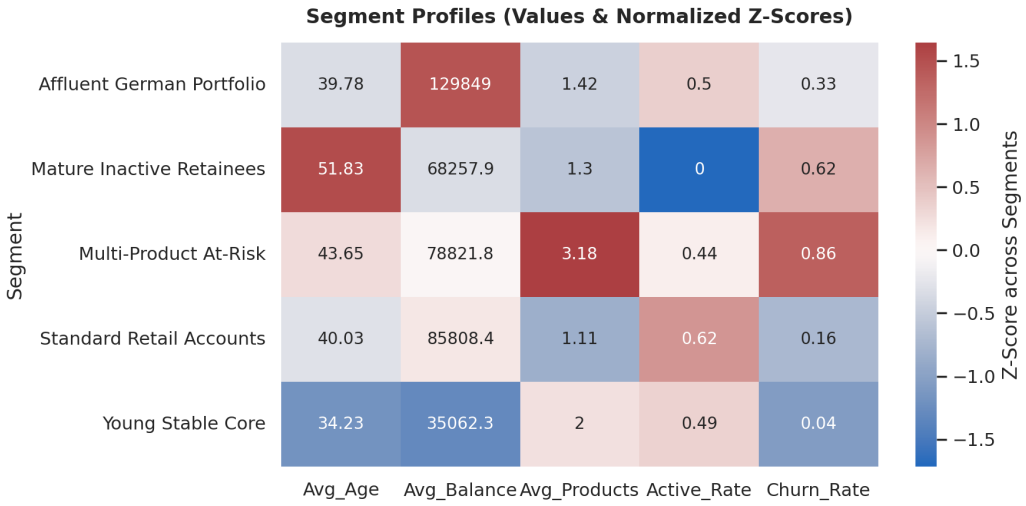


Figure 6. Segment profile matrix: raw averages and normalized Z-scores across designated customer segments.

Segment Name	Clients	Avg Age	Avg Balance	Products	Active %	Churn %
Affluent German Portfolio	1,864	39.8	€129,849	1.42	50.0%	32.8%
Mature Inactive Retainees	578	51.8	€68,258	1.30	0.0%	62.1%
Multi-Product At-Risk	326	43.7	€78,822	3.18	43.6%	85.9%
Standard Retail Accounts	4,022	40.0	€85,808	1.11	62.0%	16.1%
Young Stable Core	3,210	34.2	€35,062	2.00	49.3%	4.3%

6. Strategic Financial Recommendations & Playbook

• **German Portfolio Defense:** Deploy dedicated wealth relationship managers to German clients holding balances above €100k; introduce competitive tiered deposit yields to stem capital flight.

• **Senior & Middle-Age Advisory Packages:** For clients aged 46–60, replace standard retail checking interfaces with consolidated family wealth, pension planning, and inheritance advisory services.

• **Product Bundle Rationalization:** Immediately audit fee schedules on customers holding 3 or 4 products. Bundle services under single-fee structures with automatic fee waivers tied to deposit balances.

• **Automated Digital Reactivation Triggers:** Establish automated marketing alerts when an active customer records no logins or card transactions over 60 days, offering targeted digital cashback incentives.

7. Conclusion & Operationalization

This end-to-end framework bridges econometric segmentation and machine learning scoring. The serialized production pipeline (churn_pipeline.pkl) powers an interactive Streamlit application enabling relationship managers to score individual accounts and run counterfactual retention simulations.